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# PROJECT DRAFT

TITTLE: HR Management application.

Repository : https://github.com/NisNisSama/HRManagementProject

OVERVIEW:

The HR management application is an application designed to manage human resource data and processes. It has as objectives :

- Store and manage information about employees
- Automate HR processes as leaves, payroll and recruitment.
- Track performance and records for the HR department

MODULES :

- Employee : information, documents.
- Payroll : salary, taxes, bonus, deduction.
- Attendance : daily attendance, leave (paid), holiday (unpaid).
- Recruitment : recruitment announcement, resume database, recruitment progression.
- User management : admin, permission and access.

FUNCTIONALITIES and ACCESS:

Employee Module:

- Employee information entry : HR
- Employee information modification : HR, EMP
- Employee information deletion : ADMIN
- Document uploading : EMP
- Document update : EMP
- Document deletion : ADMIN

Payroll Module:

- Monthly salary budget report : HR overall report, HR EMP personal report
- Net Salary calculation
- Salary updating : HR
- Bonus assignation : HR

Attendance Module:

- Daily attendance sign-in : EMP
- Monthly attendance report : HR overall report, HR EMP personal report

- Leave request : EMP
- Holiday request : EMP
- Leave acceptance : HR
- Holiday acceptance : HR

Recruitment Module:

- Job listing : HR
- Job update : HR
- Job application : Public
- Resume list : HR
- Candidate selection : HR
- Recruitment progression update : HR

User Management Module:

- Roles assignation : ADMIN
- Password modification : ADMIN, EMP
- Login : EMP

TEST :

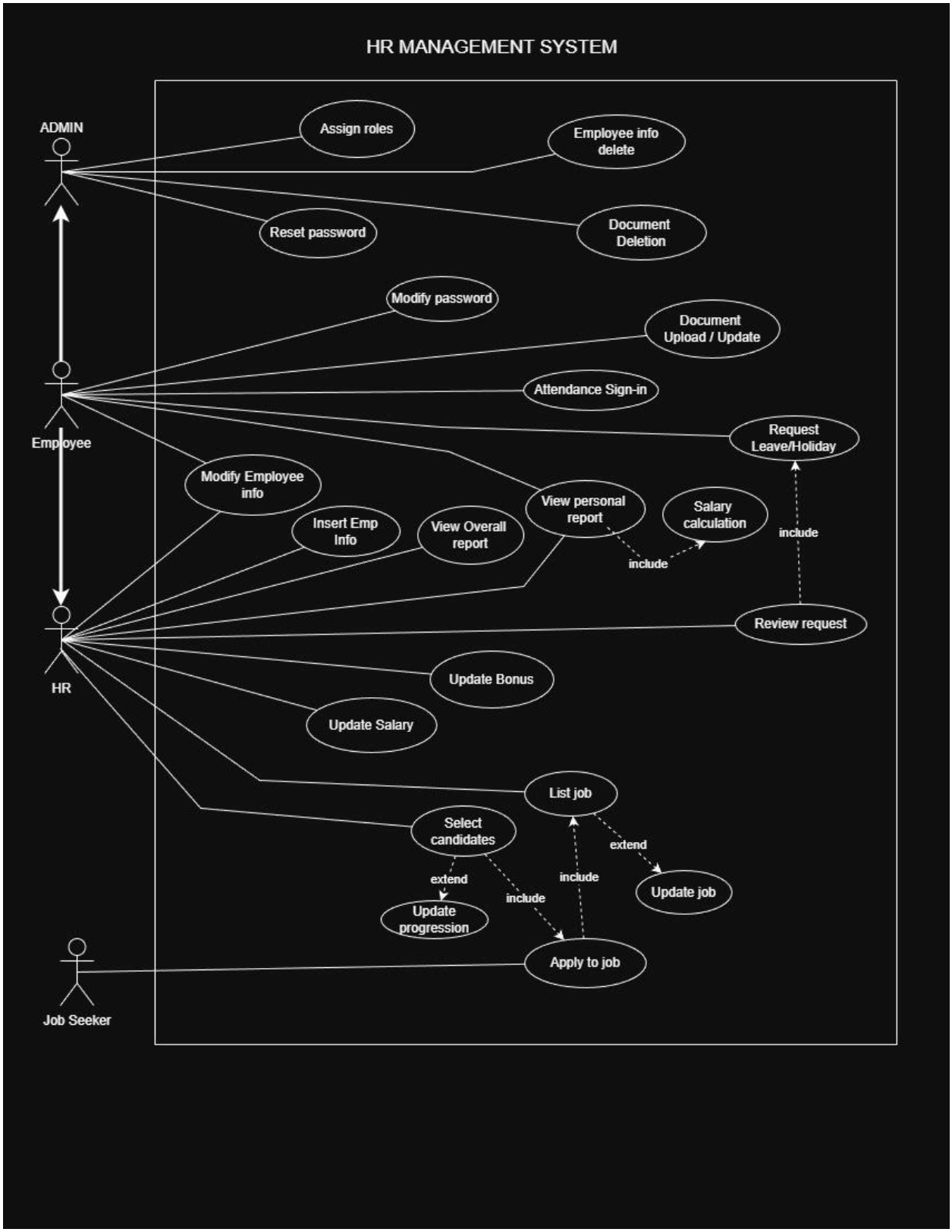
Each functionalities shall be tested one by one to ensure correct and smooth operation of the application. Listing but not limited to the following:

- Employee : login, CRUD, file upload.
- Payroll : salary calculation, edge cases.
- Attendance: daily track, leave calculation, conflict leave test.
- Recruitment : job posting creation, application submission, status tracking
- User and role management : login, logout, role based access test, permission escalation prevention.

SECURITY:

- Authentication: login
- Access management: roles and admin
- Input validation
- File format checking
- Scripting prevention

USE CASE DIAGRAM:

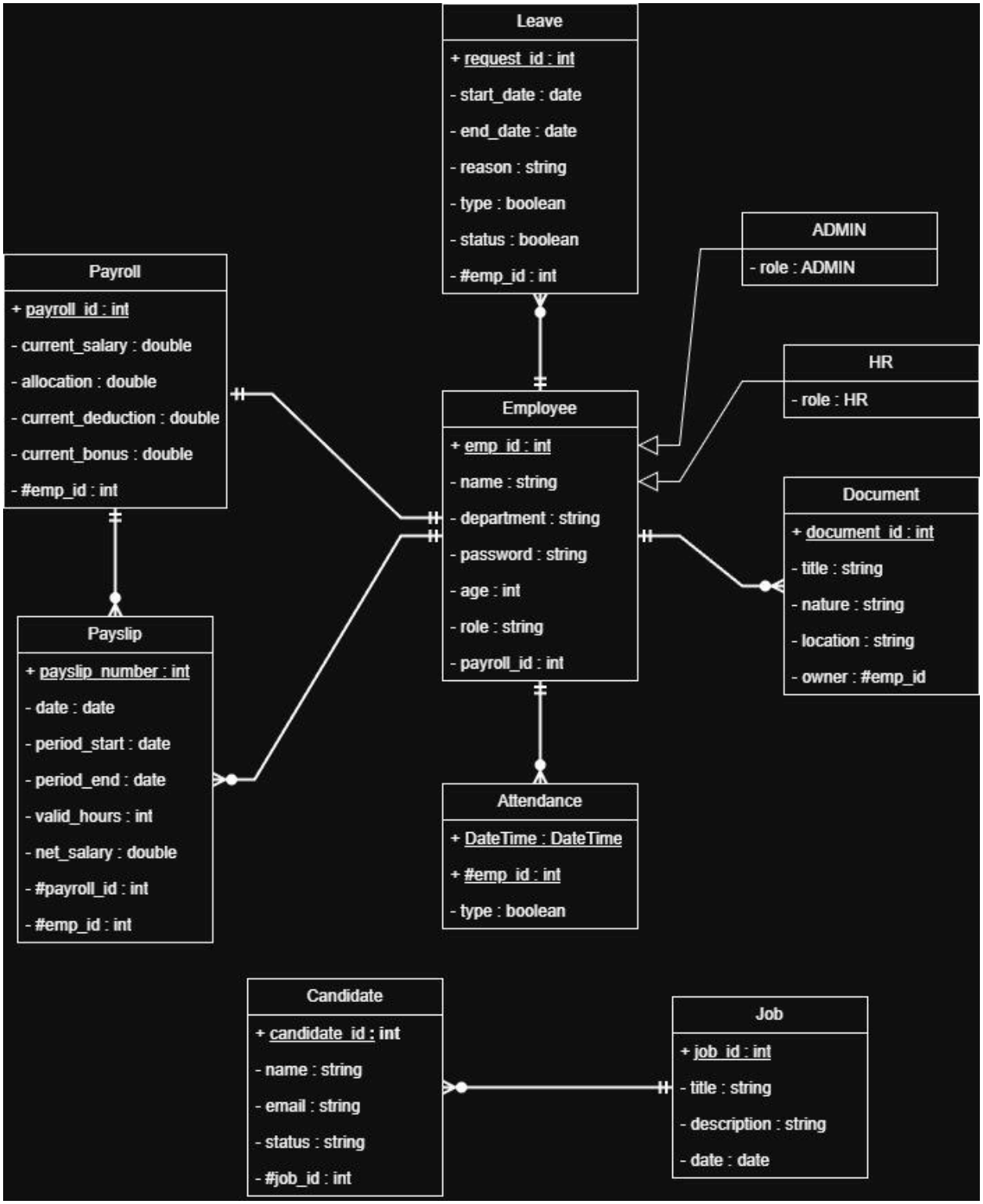

HR MANAGEMENT SYSTEM
ADMIN
Assign roles
Employee info delete
Reset password
Document Deletion
Modify password
Document Upload / Update
Attendance Sign-in
Request Leave/Holiday
Employee
Modify Employee info
View personal report
Salary calculation
Insert Emp Info
View Overall report
include
include
Review request
HR
Update Bonus
Update Salary
List job
Select candidates
extend
extend
include
include
Update job
Update progression
Apply to job
Job Seeker

CLASS DIAGRAM (PROTORYPE) :

Data Dictionary:

- Employee:
  **int** emp_id(primary_key), **string** name, **string** department, **string** password, **int** age, **char** gender, **string** role, **int** payroll_id(FK from payroll)
- Document:
  **int** document_id(primary_key), **string** title, **string** nature, **string** location, **int** owner (foreign_key from employee)
- Attendance:
  **int** DateTime+#emp_id(primary_key), **bool** type(0=out, 1=in)
- Leaves:
  **int** request_id(primary_key), **date** start_date, **date** end_date, **string** reason, **bool** type (1 = Paid, 0 =Unpaid), **bool** status (0=declined, 1=accepted, empty=pending), #emp_id
- Payroll:
  **int** payroll_id(PK), **double** current_salary, **double** allocation, **double** current_deduction, **double** current_bonus, #emp_id
- Payslip:
  **int** payslip_number(PK), **date** date, **date** period_start, **date** period_end, **int** valid_hours, **double** net_salary, #payroll, #emp_id
- Job: **int** job_id(PK), **string** title, **string** description, **date** date
- Candidate: **int** candidate_id(PK), **string** name, **string** email, **string** status, #job_id

Diagram:


Leave
+ request_id : int
- start_date : date
- end_date : date
- reason : string
- type : boolean
- status : boolean
- #emp_id : int
ADMIN
- role : ADMIN
Payroll
+ payroll_id : int
- current_salary : double
- allocation : double
- current_deduction : double
- current_bonus : double
- #emp_id : int
HR
- role : HR
Employee
+ emp_id : int
- name : string
- department : string
- password : string
- age : int
- role : string
- payroll_id : int
Document
+ document_id : int
- title : string
- nature : string
- location : string
- owner : #emp_id
Payslip
+ payslip_number : int
- date : date
- period_start : date
- period_end : date
- valid_hours : int
- net_salary : double
- #payroll_id : int
- #emp_id : int
Attendance
+ DateTime : DateTime
+ #emp_id : int
- type : boolean
Candidate
+ candidate_id : int
- name : string
- email : string
- status : string
- #job_id : int
Job
+ job_id : int
- title : string
- description : string
- date : date

SQL Script to scheme the database:

```
-- ========================================================
-- 1. INDEPENDENT TABLES (No Foreign Keys)
-- ========================================================

CREATE TABLE Job (
    job_id INT PRIMARY KEY,
    title VARCHAR(255) NOT NULL,
    description TEXT,
    date DATE NOT NULL
);

-- ========================================================
-- 2. CORE ENTITIES (Referencing Independent Tables)
-- ========================================================

CREATE TABLE Employee (
    emp_id INT PRIMARY KEY,
    name VARCHAR(255) NOT NULL,
    department VARCHAR(255) NOT NULL,
    password VARCHAR(255) NOT NULL, -- Ensure this is long enough for hashing
    age INT,
    gender CHAR(1),
    role VARCHAR(50),
      payroll_id  INT  -- Kept per diagram, nullable to avoid circular insert
deadlocks
);

CREATE TABLE Candidate (
    candidate_id INT PRIMARY KEY,
    name VARCHAR(255) NOT NULL,
    email VARCHAR(255) UNIQUE NOT NULL,
    status VARCHAR(50) NOT NULL,
    job_id INT NOT NULL,
    FOREIGN KEY (job_id) REFERENCES Job(job_id) ON DELETE CASCADE
);

-- ========================================================
-- 3. EMPLOYEE DEPENDENT TABLES
-- ========================================================

CREATE TABLE Leaves (
    request_id INT PRIMARY KEY,
    start_date DATE NOT NULL,
    end_date DATE NOT NULL,
    reason TEXT,
    type BOOLEAN NOT NULL,   -- 1 = Paid, 0 = Unpaid
    status BOOLEAN,          -- 0 = Declined, 1 = Accepted, NULL = Pending
    emp_id INT NOT NULL,
    FOREIGN KEY (emp_id) REFERENCES Employee(emp_id) ON DELETE CASCADE
);

CREATE TABLE Document (
    document_id INT PRIMARY KEY,
    title VARCHAR(255) NOT NULL,
    nature VARCHAR(100),
```

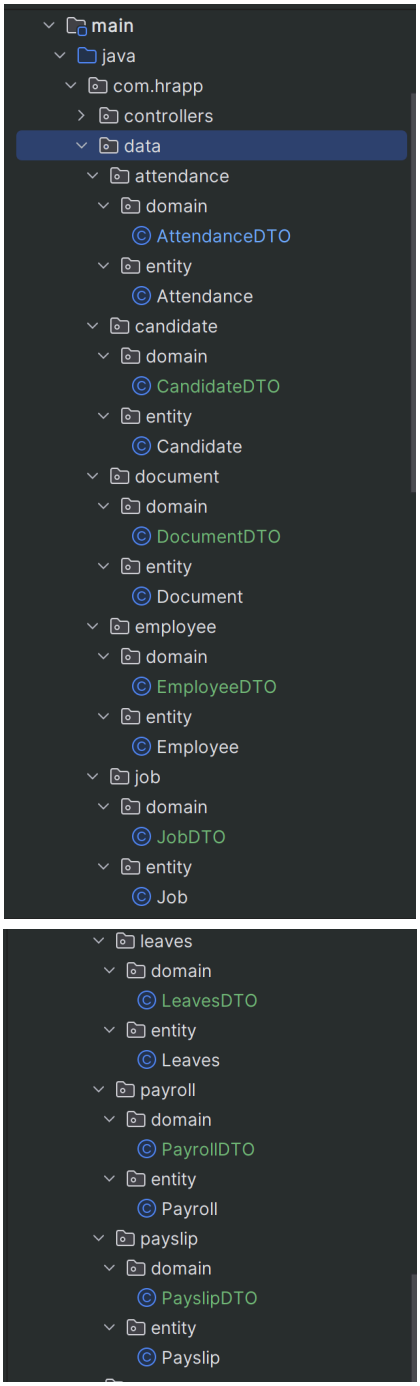
```
    location VARCHAR(512),   -- Path or URL to file storage
    owner INT NOT NULL,      -- #emp_id mapping
    FOREIGN KEY (owner) REFERENCES Employee(emp_id) ON DELETE CASCADE
);

CREATE TABLE Attendance (
    clock_time TIMESTAMP,    -- Maps to your 'DateTime' type
    emp_id INT,
    type BOOLEAN NOT NULL,   -- 0 = Out, 1 = In
    PRIMARY KEY (clock_time, emp_id), -- Composite Primary Key
    FOREIGN KEY (emp_id) REFERENCES Employee(emp_id) ON DELETE CASCADE
);

-- ============================================================
-- 4. PAYROLL & FINANCIAL ENTITIES
-- ============================================================

CREATE TABLE Payroll (
    payroll_id INT PRIMARY KEY,
    current_salary DOUBLE PRECISION NOT NULL,
    allocation DOUBLE PRECISION DEFAULT 0.0,
    current_deduction DOUBLE PRECISION DEFAULT 0.0,
    current_bonus DOUBLE PRECISION DEFAULT 0.0,
    emp_id INT NOT NULL,
    FOREIGN KEY (emp_id) REFERENCES Employee(emp_id) ON DELETE CASCADE
);

CREATE TABLE Payslip (
    payslip_number INT PRIMARY KEY,
    date DATE NOT NULL,
    period_start DATE NOT NULL,
    period_end DATE NOT NULL,
    valid_hours INT NOT NULL,
    net_salary DOUBLE PRECISION NOT NULL,
    payroll_id INT NOT NULL,
    emp_id INT NOT NULL,
    FOREIGN KEY (payroll_id) REFERENCES Payroll(payroll_id) ON DELETE CASCADE,
    FOREIGN KEY (emp_id) REFERENCES Employee(emp_id) ON DELETE CASCADE
);

-- ============================================================
-- 5. RESOLVING THE CIRCULAR RELATIONSHIP
-- ============================================================
-- Because Employee references Payroll and Payroll references Employee,
-- we add the Foreign Key constraint to Employee *after* both tables exist.

ALTER TABLE Employee
ADD CONSTRAINT fk_employee_payroll
FOREIGN KEY (payroll_id) REFERENCES Payroll(payroll_id) ON DELETE SET NULL;
```

BACKEND AS MICROSERVICE USING MICRONAUT
IMPLEMENTATION WITH MICRONAUT

Domains definition:

We will map the tables above to our micronaut application. It is important to note that this is only the representation of the database in our application, the payload used for the api calls will be under the DTO section.

Here is our folder structure after the implementation of domains



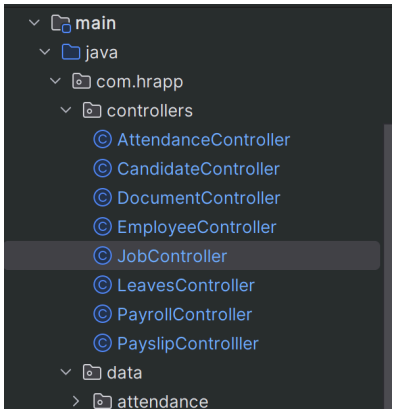
Notice that our application directories are separated following Micronaut flow logic. As DTO and Domain contain no business logic or any processing, they are just pure data that will be manipulated by the application. This is why it is isolated from the rest of the logic implementation.

Controller Initialization

The controllers of each module must be initialized with all the basic methods for the endpoint. GET ONE, GET ALL, CREATE, UPDATE and DELETE.

```java
@Controller("/job")  no usages  Windows-Nisaina *
public class JobController {
    @Get("/{jobId}")  no usages  new *
    public JobDTO oneJob(Integer jobId){
        JobDTO job = null;
        return job;
    }

    @Get("/all")  no usages  new *
    public List<JobDTO> allJob(){
        List<JobDTO> jobList = List.of();
        return jobList;
    }

    @Post("/create")  no usages  new *
    public String createJob(Integer jobId, String title, String description, LocalDate date){
        new JobDTO(jobId, title, description, date);
        return "Ok";
    }

    @Post("/update/{jobId}")  no usages  new *
    public String updateJob(Integer jobId, String title, String description, LocalDate date){
        new JobDTO(jobId, title, description, date);
        return "Updated";
    }

    @Get("/delete/{jobId}")  no usages  new *
    public String deleteJob(Integer jobId) { return "Deleted"; }
}
```

This will be done to all the controllers before implementing business logic for security and filtration. These are all the files created.



DYNAMIC FRONTEND USING JAVA SERVER PAGES WITH TOMCAT 9.0

JSP will allow us to directly manage the session for each user instance of our application. This will facilitate the authentication and authorization management separating completely our HR management aspect, the Employee services aspect and the public accessible aspect for all modules.

SECURITY IMPLEMENTATION

To ensure data integrity and data security, we will have two layers of verification. First will be in the frontend using session attributes to hide/show functionalities to intended users only. It is notable that session attributes can be manipulated, so secondly, we will implement a zero trust architecture in our backend.

Consequently, for each api call hitting our endpoint, we will make an identity verification for all payloads trying to access our services. This will ensure that the data will only be accessible to the intended user only.